

Doc 2 — Field Packet

PRINT THIS. Carry it on a clipboard with a pen. Plainsboro Public Library, 28 Aug 2026. Fill in every table as you go — you will not remember it later.

Header — fill in on arrival

Date / arrival time	Departure time	Weather	Surveyor
			Tarun Cherukuri
Occupancy at arrival (rough headcount)	Occupancy at peak	Device (phone model)	NetSpot version

PART A — Tonight, before you sleep

A1. Phone configuration (do not skip — these silently ruin surveys)

- ☐ **Location Services ON**, and NetSpot granted **Precise location**. Android returns zero scan results without this.
- ☐ **WiFi scan throttling OFF**. Settings → About Phone → tap Build Number 7× → back to Developer Options → toggle 'WiFi scan throttling' off. Otherwise Android caps you at ~4 scans per 2 minutes and you will collect duplicated garbage across the whole building.
- ☐ **Battery optimization OFF** for NetSpot (Settings → Apps → NetSpot → Battery → Unrestricted).
- ☐ **Do Not Disturb ON** for tomorrow's walk. Your screenshots showed an active video call overlay sitting on top of the app — that will cover the UI and cost you readings.
- ☐ **Auto-rotate OFF**, screen timeout set to 5+ minutes.
- ☐ Charge to 100%. Pack a power bank and cable.

A2. CRITICAL — your project is contaminated. Fix it tonight.

Project 1 currently holds **6 data points from your apartment, plotted onto the library's floor plan**. The two AP markers on that map are labelled Verizon_7XFC94 — your home router. If you tap 'Resume scanning' at the library tomorrow, those apartment readings blend into your library dataset and the whole survey is worthless.

The fix, and it is easy: a Zone can hold multiple *snapshots*, and each snapshot is an independent dataset on the same floor plan and scale. Tomorrow you will use **Zone** → ... → **Add new snapshot** to start a clean one, not 'Resume scanning'. Delete the practice snapshot (#1 Aug 27) tonight so there is no chance of tapping into the wrong dataset while you are standing in the library.

- ☐ Delete snapshot '#1 Aug 27, 2026' from Zone 1 (it holds the apartment data).
- ☐ Rename the project: **Project 1** → **Plainsboro-Library-2026-08-28** (project menu → three dots).
- ☐ Rename the zone: **Zone 1** → **Children's Level** (Zone list → ... → Rename zone).
- ☐ Add a **Zone description** while you are there — note the floor plan source and its date. Future you will want to know where that drawing came from.
- ☐ Delete the blank white 'Project 2'. Keep 'Sample — Office Space' as your Planning reference.

A2b. Verify the scale is actually calibrated

The practice snapshot was dropped onto the plan quickly, and there is a real chance the scale was never set. An uncalibrated project still produces a colorful heatmap — it just produces one where every distance is meaningless.

- ☐ Open the zone and confirm a scale is set. If not, calibrate it tomorrow before your first point (Part C2).
- ☐ The floor plan's own KEY and room numbering give you reference features to measure against.

A3. You now have the whole building — all three floors

The Architectural Record set (published 2011) gives you Ground, Second, and Third (Children's) level plans. Save all three images to your phone tonight. Two things on those drawings are worth more than they look:

1. Every plan has a graphic scale bar (the '0 — 10FT' marker) **and a north arrow.** The scale bar means you can calibrate without pacing anything: tap the two ends of the bar in NetSpot and enter 10 ft. Because it is a short reference, tap precision matters — zoom the image right in before tapping, then sanity-check by measuring one real room dimension on site and comparing. The north arrow lets you orient each zone consistently and describe locations compass-wise in the report instead of 'the top left bit'.

2. These are 2011 design drawings, not as-builts. 'Future Aquarium' is the tell. Fifteen years of a working library means furniture has moved, rooms have been repurposed, and shelving has been added. Verify as you walk and annotate every difference — a documented drawing-vs-reality discrepancy list is a legitimate finding in its own right and shows you don't take inputs on trust.

- ☐ Save all three floor plan images to the phone at the highest resolution you can get.
- ☐ Credit the source in your report: floor plans from Architectural Record's 2011 feature on the building. Attribution costs a line and protects you.
- ☐ Note that the plan set is a design record, so square footage and room labels may not match today's building.

A3b. Zone plan — what to expect on each floor

Zone	From the plan key	Your RF priorities there
Ground (Level 1)	Entry, Main Circulation Desk, Cafe, Main Reading Room, Lounge, Newspaper Shelves, Copy/Print/Fax, Used Book Sales, Book Sorting Room, Computer Workstations, Gallery for Local Artists, Stacks, Periodicals, Solitary Study Desks, Community Room, Tech Support, Reading Garden	Community Room — the big seated space; this is your capacity story, not a coverage one. Computer Workstations — client density. Stacks and Periodicals — paper attenuation. Reading Garden — outdoor; measure signal leakage past the building envelope. Cafe — check for a microwave (2.4 GHz noise source).
Second	Stacks, Lounge, Local History / Conference Room, Reference Desk / Office, Staff Offices / Break Room, six Group Study Rooms, Health & Wellness Consultation Room, Mechanical Room	Group Study Rooms — six enclosed boxes in a row; take a point inside and outside each door. This is the single best attenuation dataset in the building. Mechanical Room — do not enter; measure the corridor beside it and watch SNR for a noise signature. Break Room — likely microwave. The large crossed square on the plan is an open-to-below void — see below.
Third (Children's)	Children's Reception Desk, Reading Oval / 'Cone of Light', Computer Bar, Future Aquarium, Science Center, Program Room, Quiet Terrace, Active Terrace, Staff Offices	Program Room — enclosed, likely the worst coverage on the floor. Computer Bar — density. Both terraces — outdoor coverage and leakage. Cone of Light — the reading oval sits under the vertical void.

The atrium is your best finding, if it exists as drawn. That large crossed-out square on the second floor plan is an open-to-below space, and the 'Cone of Light' suggests it runs the full height. A vertical void is an RF chimney: signal from one floor's APs propagates freely to the others through it, while the solid floor slabs elsewhere block it. Take points at the void edge on every level and compare. If you find an AP on one floor serving clients on another through the atrium — and note what that does to roaming and co-channel contention — you have written something most junior candidates would never think to look for.

A3c. Realistic time budget

Phase	Time	Notes
Arrival, staff conversation, baseline Inspector scan	20 min	Part B
Project and zone setup, three plans calibrated	20 min	Do all three up front
Ground floor walk	45 min	Largest public footprint
Second floor walk	40 min	Study rooms take time — two points each
Third floor walk	40 min	You already have this plan loaded
Exit procedure	15 min	Part E — do not compress this
Total	~3 hours	Plus travel. Eat before you go.

Recommendation: survey all three floors at ~40 points each rather than two floors at 70. A whole-building survey with an honestly stated density is worth more than a partial one at higher resolution, and the interesting story here — the vertical propagation through the atrium — only exists if you have all three levels. If you run out of time or energy, stop cleanly and say so in the report; do not fake the last floor with six scattered points.

A4. Calibrate yourself

- ☐ Measure a known distance at home (a hallway, 15–20 ft). Walk it normally, count your steps, divide. Write your pace length here: _____ **inches per step**. You will use this to scale the floor plan and to place points accurately.
- ☐ Do one 10-point practice survey in your apartment on a loaded floor plan. Watch the 'map coverage' number climb. Learn what the tap-and-wait rhythm feels like before you are standing in public.

A5. Tune the heatmap display settings

These are **display** settings, not capture settings — they change how collected data is rendered, so you can adjust them afterward and re-render. Set them tonight, then leave them alone in the library.

Setting	Default	Set it to	Why
Min signal level	–96 dBm	–85 dBm	Real indoor readings cluster between about –35 and –85. At full range, 90% of the color gradient goes unused and everything renders as one flat wash of orange — exactly what your practice heatmap looked like.
Max signal level	–10 dBm	–35 dBm	Same reason. Narrowing the range makes weak and strong areas visually distinct.
AP detection sensitivity	–40 dBm	Start at –40; drop toward –65 if needed	This is the threshold for NetSpot estimating an AP's position. At –40 it only places markers for radios you practically stood under. If it shows fewer APs than you physically counted, lower it and re-render. Too low and it starts inventing markers for distant neighbours.
Gradient type	Blue→red	Keep	The default rainbow gradient is the one wireless people read fastest.
Survey path	On	Keep ON	Draws your walk trail on the map. It is visible evidence of your methodology — genuinely worth including in the report.
Gray out unmeasured area	On	Keep ON	This is the honest setting. Turning it off lets a sparse survey look complete when it isn't. Leave it on and let the gray tell the truth.
Sampling points	Simple	Keep	Display style for point markers. No effect on data.

A6. Pack list

- ☐ Phone (the instrument) + power bank + cable
- ☐ This packet, printed, on a clipboard
- ☐ Two pens
- ☐ Tape measure or laser measure if you own one
- ☐ iPad — **notes and backup camera only**. NetSpot on iOS cannot read signal strength; it only builds speed-based heatmaps. It is not a survey device.
- ☐ Student ID (helps when you explain yourself)
- ☐ Water. This takes 1–2 hours on your feet.

PART B — Arrival (first 10 minutes)

B1. Talk to the front desk first

Say roughly this: "Hi — I'm a grad student in cybersecurity and network engineering at UMD. I'm practicing WiFi coverage mapping for a portfolio project. I'd just be walking around with my phone taking signal readings on the public WiFi — nothing that touches your systems. Is that okay?"

Then ask these three, and write the answers down:

Question	Answer
Do you have a floor plan I could photograph? (esp. the adult/main level)	
Do you know roughly where the access points are mounted?	
Are there areas where staff or patrons complain the WiFi is bad?	
Any areas I should not walk into?	
Name / role of the person you spoke to	

Scope boundary — hold this line. You are listening to broadcast beacons and nothing else. No deauthentication, no packet capture of other patrons, no connecting to anything you were not offered, no attempts to reach management interfaces. Passive RF measurement is entirely legitimate; the moment you cross into interacting with their infrastructure it stops being a portfolio project.

B2. Baseline scan

- ☐ Connect to the public WiFi, accept any captive portal.
- ☐ Open **Inspector** → **Networks**. Screenshot the full list, scrolling to capture everything.
- ☐ Open **Inspector** → **Channels**, select **2.4 GHz**. Screenshot.
- ☐ Switch to **5 GHz**. Screenshot.
- ☐ Open **Inspector** → **Comparison**, let it run ~60 seconds near the entrance. Screenshot.
- ☐ Fill in Table 1 and Table 2 below from what you see.

Table 1 — SSID inventory (the library's own networks)

SSID	BSSID (last 6)	Band	Channel (width)	RSSI here	Security	Vendor

SSID	BSSID (last 6)	Band	Channel (width)	RSSI here	Security	Vendor

Table 2 — Channel usage at the entrance

Record how many APs NetSpot sees on each channel, and its rating. This is your co-channel interference evidence.

2.4 GHz ch	# APs	Rating	5 GHz ch	# APs	Rating
1			36		
2			40		
3			44		
4			48		
5			52		
6			56		
7			60		
8			64		
9			100		
10			104–144		
11			149		
Other			153–165		

PART C — Floor plan and project setup

C1. If you need to sketch a level

- ☐ Walk the perimeter of the public area once, just looking.
- ☐ Sketch a top-down outline: exterior walls, interior walls, stacks, study rooms, restrooms, stairs, elevator, entry.
- ☐ Pick one long straight wall and pace it. Write the distance on the sketch. That is your calibration reference.
- ☐ Photograph the sketch flat, from directly above, no shadow. Rough and to-scale beats pretty and wrong.

C2. NetSpot project setup — snapshot workflow

- ☐ Open **Plainsboro-Library-2026-08-28**.
- ☐ Zone **Children's Level** → ... → **Add new snapshot**. Do NOT tap 'Resume scanning' — that would append to an existing dataset.
- ☐ Confirm the floor plan is loaded and oriented the way you are physically standing. Rotate yourself, not the map.
- ☐ **Calibrate the scale**: tap two points whose real separation you know, enter the distance. Every distance calculation downstream depends on this. Use the longest reference you can measure — a short reference multiplies its own error across the whole plan.
- ☐ Create all three zones up front: **Zone list** → + for Ground and Second, each with its own plan image and its own scale calibration. Doing this before you start walking saves fumbling later.
- ☐ **Calibrate from the drawing's scale bar**: zoom the image in hard, tap the two ends of the '0—10FT' marker, enter 10 ft. Then verify against one real measured dimension on site before you trust it.
- ☐ Name zones plainly: 'Children's Level', 'Main Level'. Never leave them as 'Zone 1'.

Snapshots are your secret weapon. Same zone, same floor plan, different moment in time. If you can manage a quiet-period walk and a busy-period walk, you get a coverage-versus-capacity comparison on identical geometry — and that is a genuinely sophisticated thing for a junior candidate to have produced. Name them by time: 'AM-quiet',

'PM-busy'.

Table 3 — Zone register

Zone name	Scale ref used	Real dist (ft)	Start	End	Points	Coverage %
Ground						
Second						
Third (Children's)						

PART D — The walk

D1. Technique — consistency is the whole game

- Hold the phone at **chest height, screen up, same orientation every single time.**
- Hold it **away from your body.** You are a bag of salt water and you attenuate 3–6 dB. Inconsistent body position = inconsistent data.
- Do not cover the top edge of the phone with your hand.
- **Stand still** while a point is being collected. Walk → stop → tap your location → wait for the scan to complete → walk again.
- Tap where you *actually are*. NetSpot has no positioning system; you are the positioning system. Locate yourself against landmarks — wall corners, shelf ends, door frames, columns.

D2. Where the points go

Priority	Placement	Why
Grid	Every 15–20 ft through open areas, snake pattern up and down aisles	Baseline coverage
Corners	Every corner of every room	Edges are where coverage fails
Doorways	Both sides of every wall and door	Captures attenuation — your best material
Stacks	Between and behind bookshelf rows	Dense paper kills signal; this is a library's signature finding
Enclosed rooms	Program room, study rooms, staff areas if permitted	Often the worst coverage in the building
Under APs	Directly beneath every AP you spot	Gives you a reference maximum
Dead zones	Extra density anywhere signal looks bad	Do not avoid bad areas — they are the point
Perimeter	Just outside exterior walls: entrance, terraces, parking edge	Signal leakage outside the building is a security finding
Vertical	Stairwells, near the elevator	Between-floor bleed and shaft shadowing

D3. What 'map coverage %' actually means — do not be fooled by it

Your 6-point practice run reported **75%, good**. Six points cannot characterize a library floor. The number measures **spatial spread** — how much of the plan sits within interpolation range of some point — not density and not accuracy. A handful of points flung to the corners will read 'good' while the map between them is almost entirely invented.

Your real target is density: 50–80 points per level, roughly one every 15–20 ft. Use map coverage only as a floor (get it above 80%), never as your finish line. Record both numbers in Table 3 and report both, so a reader can judge the survey's weight for themselves. Gray areas on the map are *unmeasured*, not dead — never describe them as coverage gaps.

D4. Use Inspector during the walk, not just at the ends

- ☐ Every 15–20 minutes, drop out to **Inspector** → **Networks** and check whether new BSSIDs have appeared that you had not seen from the entrance. That is how you discover APs in parts of the building you have just reached.
- ☐ Whenever a reading surprises you, switch to Inspector on the spot and look at *which* BSSID is serving you and on what channel. The heatmap tells you 'weak here'; Inspector tells you *why*.
- ☐ In any area that feels bad, check **Channels** for both bands right there. Channel congestion is local — the entrance reading does not describe the whole building.

☐ Log every new BSSID into Table 1 as you find it. Distinct BSSIDs on the same SSID = your AP count.

Table 4 — Access point inventory (fill as you spot them)

Mark each AP on your printed/sketched floor plan with a circle and the same AP number.

AP #	Location description	Mount (ceiling/wall/shelf)	Est. height	Model / vendor if visible	Notes

Table 5 — Observations log

Time	Location	Observation	Suspected cause

Table 6 — Construction and obstacles

Area	Wall / material type	Est. attenuation impact	Evidence (RSSI before / after)

Table 7 — Interference sources spotted

Microwaves, cordless phones, Bluetooth speakers, wireless presentation gear, security cameras, baby monitors, neighbouring businesses' APs, motorized equipment.

Source	Location	Band affected	Notes

Table 8 — Active spot checks (speed tests)

Run 4–6 of these at deliberately chosen spots: right under an AP, mid-room, at the worst location you found, and at the building edge.

#	Location	RSSI at spot	Download	Upload	Latency	Time

Table 9 — Occupancy / time log

Client load changes RF conditions. Note it so your results are interpretable.

Time	Area	Approx. people present	Notable activity

Time	Area	Approx. people present	Notable activity

PART E — Before you walk out the door

Do not skip this. A second trip costs you an hour of driving; these checks cost ten minutes.

E1. Quality check while you can still fix it

- ☐ Open the heatmap and **look at it**. Obvious blank regions? Go back and fill them now.
- ☐ Record point count AND map coverage % for every zone in Table 3. Under 50 points or under 80% coverage means go back out.
- ☐ Walk to the two or three worst-looking areas and add extra points. Detail where it matters.

E2. Filter the heatmap to the library's networks

The map menu (three dots, top right) has **Include/exclude networks**. This is the most under-used feature in the app and it is what separates a clean deliverable from a mess.

- ☐ Deselect every neighbouring network. Render the heatmap showing **only the library's SSIDs** — this is your primary coverage map.
- ☐ Then render **one SSID at a time** if they run separate staff and guest networks. They may have different coverage.
- ☐ Then render **only the neighbours**. That map is your external interference exhibit, and almost nobody at your level thinks to produce it.
- ☐ Screenshot each version. Label them as you go so you can tell them apart tonight.

E3. Capture everything

- ☐ Cycle all three heatmap types — **Signal level, SNR, SIR** — per zone, per filter. Use **Export to image** from the map menu rather than a screenshot: cleaner output, no status bar, report-ready.
- ☐ **Save the project.**
- ☐ **Export project** (.nsp file) from the project screen.
- ☐ **Email the export to yourself or push it to cloud storage before you leave the building.** Never let it exist in exactly one place.
- ☐ Re-run Inspector → Networks, Channels (2.4 and 5), Comparison. Screenshot. You now have entrance-versus-interior comparison data.
- ☐ Run your remaining speed tests (Table 8) and record them.
- ☐ Photograph every AP you could see, plus general shots of the space.
- ☐ Photograph your completed paper tables in this packet as a backup.
- ☐ Confirm every table in this packet has something written in it.
- ☐ Thank the staff member you spoke to. Offer them a copy of the findings.

PART F — Same evening, while it is fresh

- ☐ Type up every paper table into Doc 3 (report template) tonight. You will not remember the detail in three days.
- ☐ Write down the three things that surprised you.
- ☐ Back up the .nsp export in a second location.
- ☐ Sketch the report outline: scope → methodology → findings → recommendations.

Quick reference — tear this corner off if you like

dBm	Verdict	SNR	Verdict
–30 to –45	Excellent	40 dB+	Excellent
–46 to –60	Very good	25–40 dB	Very good
–61 to –67	Good (design floor)	15–25 dB	Fair
–68 to –72	Marginal	10–15 dB	Poor
–73 to –80	Poor	Under 10 dB	Unusable
Below –80	Unusable	—	—

2.4 GHz non-overlapping channels: **1, 6, 11**. Design target: **–67 dBm or better, SNR > 20 dB**, cell overlap 15–20%.